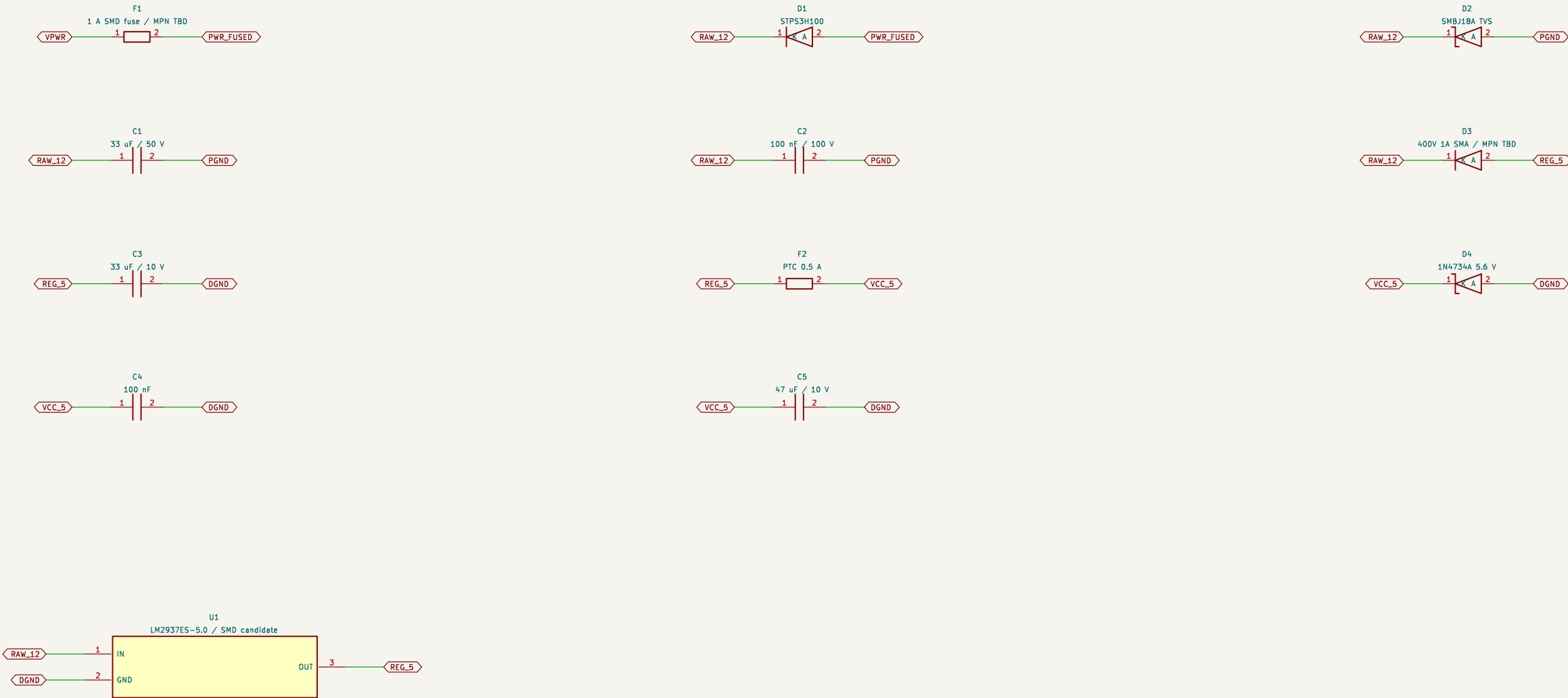


1997 GRAND MARQUIS MS3 CUSTOM ECU							
Genuine MS3 daughtercard + custom mainboard/MS3X functions + external MicroSquirt transmission control							
REV A1: COMPONENT-LEVEL ENGINEERING DRAFT. NOT A FABRICATION RELEASE.							
A	01 Switched power and 5 V regulation		10 Factory crank VR conditioner		19 Low-side auxiliary drivers – group 3		
	Switched power and 5 V regulation		Factory crank VR conditioner		Low-side auxiliary drivers – group 3		
	File: 01_power.kicad_sch		File: 10_crank_vr.kicad_sch		File: 19_aux_3.kicad_sch		
B	02 Ground topology, reference rails and CAN		11 Crank adjustment, phase selection and cam front end		20 Oxygen-sensor heater low-side stages		
	Ground topology, reference rails and CAN		Crank adjustment, phase selection and cam front end		Oxygen-sensor heater low-side stages		
	File: 02_ground_can.kicad_sch		File: 11_crank_cam_adjust.kicad_sch		File: 20_oxygen_heaters.kicad_sch		
C	03 Genuine MS3 daughtercard – 40-pin base socket		12 Cam zero-crossing comparator		21 Protected switch inputs		
	Genuine MS3 daughtercard – 40-pin base socket		Cam zero-crossing comparator		Protected switch inputs		
	File: 03_ms3_base.kicad_sch		File: 12_cam_comparator.kicad_sch		File: 21_digital_inputs.kicad_sch		
D	04 Genuine MS3 expansion interfaces		13 Sequential injectors A–D		22 Brake, A/C request and flex-fuel input		
	Genuine MS3 expansion interfaces		Sequential injectors A–D		Brake, A/C request and flex-fuel input		
	File: 04_ms3_expansion.kicad_sch		File: 13_injectors_1.kicad_sch		File: 22_vehicle_switches.kicad_sch		
E	05 Factory Ford connector – cavities 1 to 52		14 Sequential injectors E–H		23 External MicroSquirt transmission interface		
	Factory Ford connector – cavities 1 to 52		Sequential injectors E–H		External MicroSquirt transmission interface		
	File: 05_ford_1.kicad_sch		File: 14_injectors_2.kicad_sch		File: 23_transmission.kicad_sch		
F	06 Factory Ford connector – cavities 53 to 104		15 Eight buffered smart-coil signals		24 Service and spare feature headers		
	Factory Ford connector – cavities 53 to 104		Eight buffered smart-coil signals		Service and spare feature headers		
	File: 06_ford_2.kicad_sch		File: 15_spark_logic.kicad_sch		File: 24_service.kicad_sch		
G	07 Engine temperature, throttle and oxygen inputs		16 Optional factory wasted-spark power stages		25 Factory functions outside the implemented control scope		
	Engine temperature, throttle and oxygen inputs		Optional factory wasted-spark power stages		Factory functions outside the implemented control scope		
	File: 07_engine_analog.kicad_sch		File: 16_factory_ignition.kicad_sch		File: 25_retention_limits.kicad_sch		
H	08 MS3X analogue inputs		17 Low-side auxiliary drivers – group 1		26 ECM-shell pass-through to cabin MicroSquirt		
	MS3X analogue inputs		Low-side auxiliary drivers – group 1		ECM-shell pass-through to cabin MicroSquirt		
	File: 08_expansion_analog.kicad_sch		File: 17_aux_1.kicad_sch		File: 26_shell_passthrough.kicad_sch		
I	09 MAP sensor, battery measurement and spare ADC		18 Low-side auxiliary drivers – group 2				
	MAP sensor, battery measurement and spare ADC		Low-side auxiliary drivers – group 2				
	File: 09_map_battery.kicad_sch		File: 18_aux_2.kicad_sch				
Global net labels carry all inter-sheet electrical connections. Open a sheet to inspect components and pin numbers. Read docs/Design_review.md and reports/validation.md before implementing or purchasing parts. This draft does not implement Ford OBD diagnostics or establish automotive electrical/thermal qualification.							
NOT FOR FABRICATION OR VEHICLE CONNECTION							
Sheet: / File: GrandMarquis97_RevA.kicad_sch							
Title: 1997 Grand Marquis – MS3 custom carrier							
Size: A3				Date: 2026-09-12			Rev: A1 REVIEW
KiCad E.D.A. kicad-cli 7.0.11+dfsg-1build4				Id: 1/27			

01 | Switched power and 5 V regulation

VPWR feeds logic only. Injector, coil and transmission load supplies remain externally fused.



POWER REVIEW: nominal 5 V / 500 mA architecture, inherited from the V3.0 approach.
At 14 V and 0.35 A the regulator dissipates about 3 W; thermal design is unresolved.
This protection circuit has not passed reverse battery, cranking, load dump or EMC testing.
Fuse, PTC and bulk-capacitor footprints remain blank pending exact part selection.

SUPPLY: vehicle 12 V through retained factory fused feeds; voltage regulation inside ECM. Existing onboard F1 is retained as a separate draft protection element. Retain factory fuse protection; input current budget and transient/reverse-polarity performance require validation.

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1997 Grand Marquis / genuine MS3

Sheet: /01 | Switched power and 5 V regulation/
File: 01_power.kicad_sch

Title: 01 | Switched power and 5 V regulation

Size: A3 Date: 2026-09-12

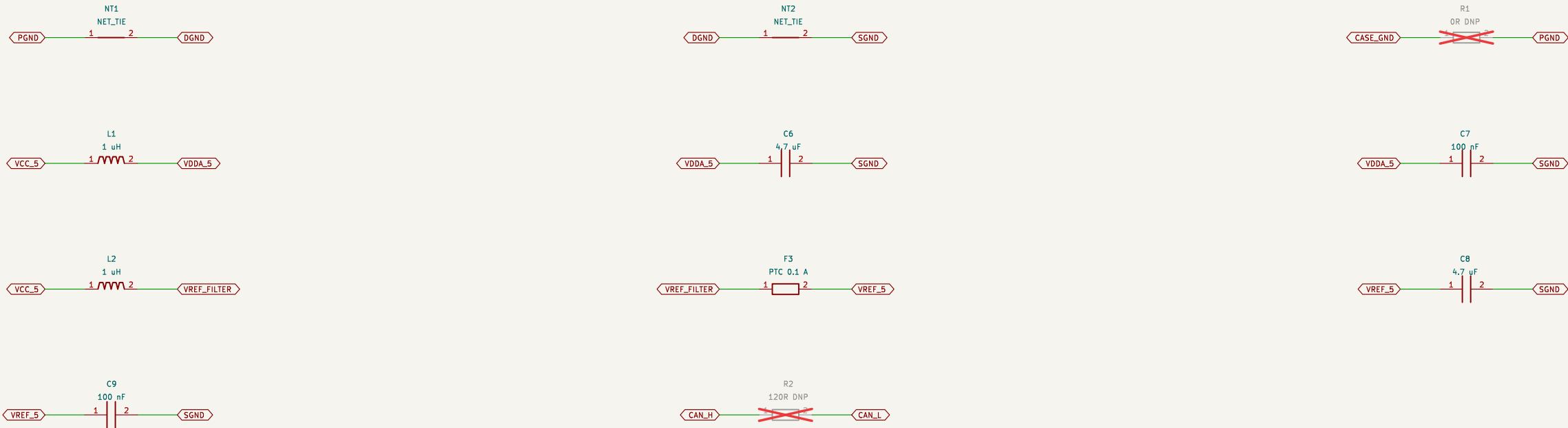
KiCad E.D.A. kicad-cli 7.0.11+dfsg-1build4

Rev: R4 CURRENT-TARGET RE

Id: 2/27

02 | Ground topology, reference rails and CAN

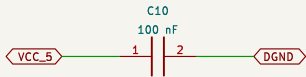
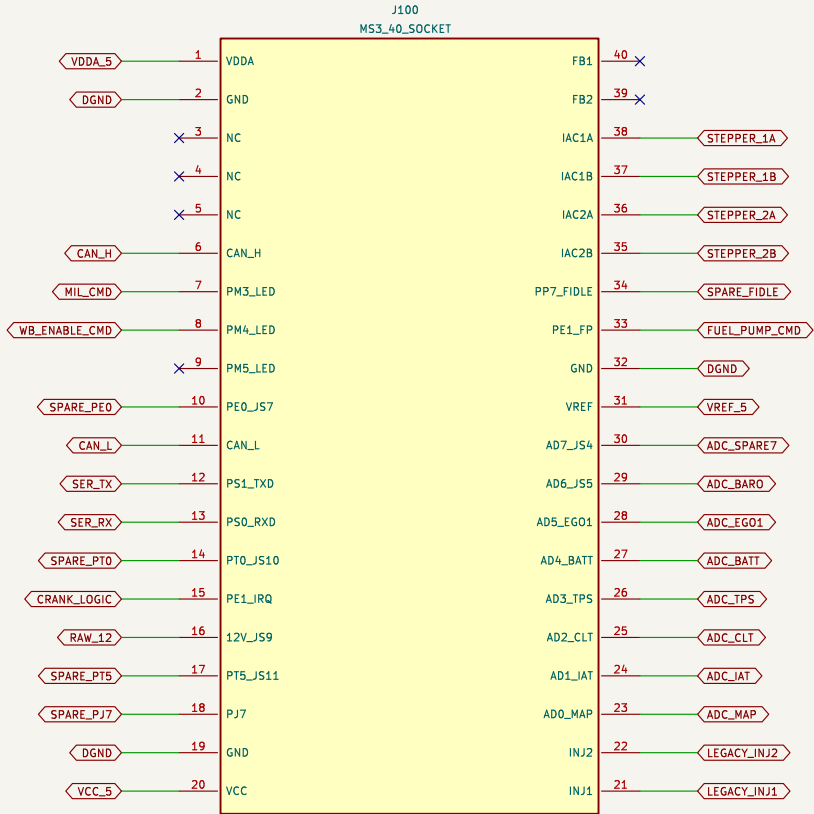
Physical star placement is required at the connector/driver return region. Net-tie values represent copper joins.



Genuine MS3 socket pins 6 and 11 carry CAN_H / CAN_L: do not add a second transceiver.
Check total bus resistance with power off; use one terminator at each end of the bus.
PGND: Ford 24, 51, 76, 77, 103. SGND: Ford 91.
Keep dirty returns physically separate until the deliberate star; connector current ratings need verification.

03 | Genuine MS3 daughtercard – 40–pin base socket

Signal assignments follow the MS3–annotated V3.0 schematic, hardware guide page 197. Genuine module internals are not recreated.



PM3 / PM4 / PM5 are repurposed LED ports: MIL, heater enable and optional EGR. These assignments require firmware output–resource review before use. USB and SD remain on the genuine card. Legacy batch injector and stepper ports are exposed on the service header; no duplicate legacy injector stages fitted. Do not substitute an MS1/MS2 pinout. The module is powered through its documented 5 V and 12 V interfaces.

NOT FOR FABRICATION OR VEHICLE CONNECTION

1997 Grand Marquis / genuine MS3

Sheet: /03 | Genuine MS3 daughtercard – 40–pin base socket/
File: 03_ms3_base.kicad_sch

Title: 03 | Genuine MS3 daughtercard – 40–pin base socket

Size: A3 Date: 2026–09–12

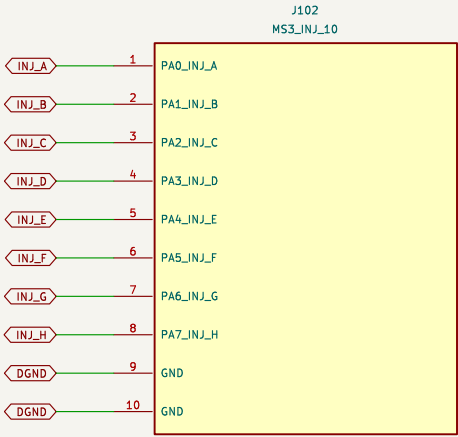
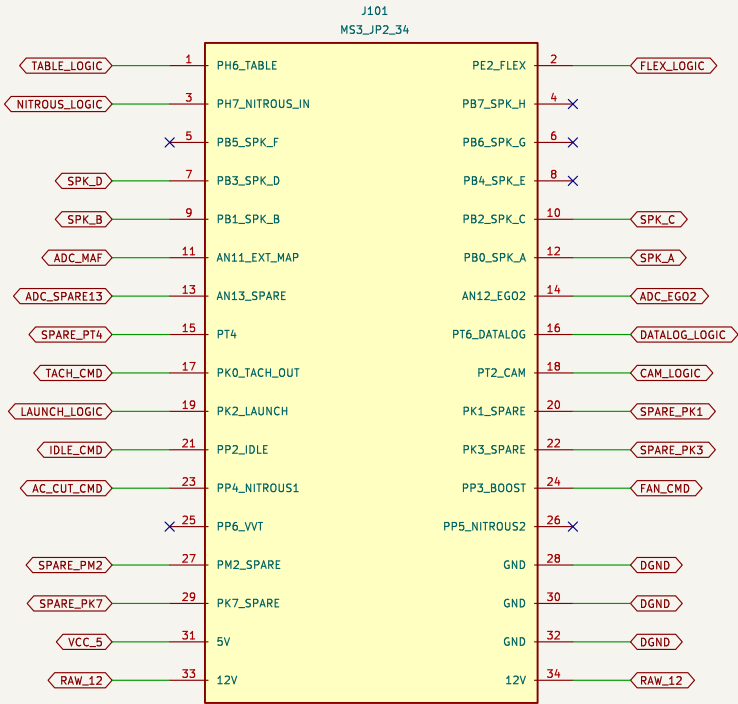
KiCad E.D.A. kicad–cli 7.0.11+dfsg–1build4

Rev: R4 CURRENT–TARGET RE

Id: 4/27

04 | Genuine MS3 expansion interfaces

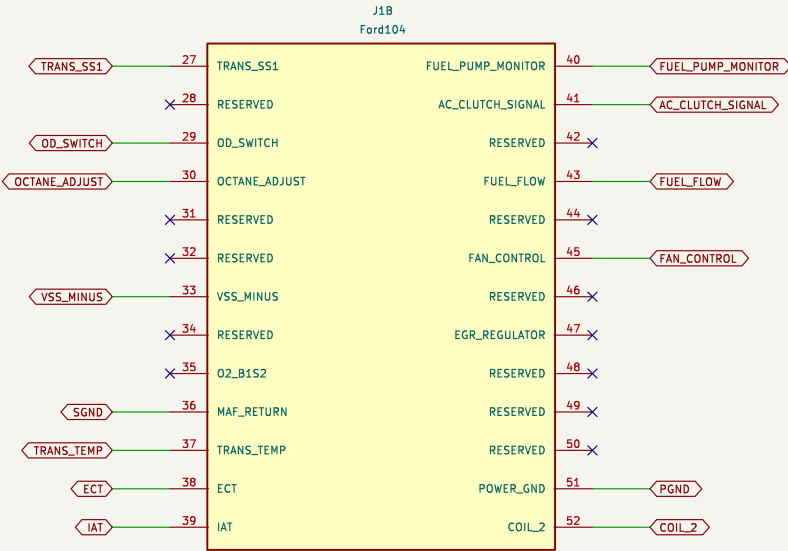
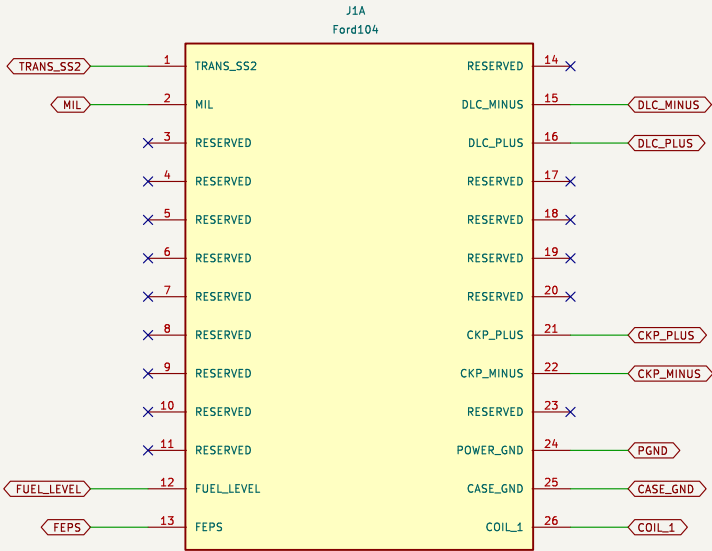
JP2 and injector header use exact numbered module pins from hardware guide page 205. These are not DB37 pin numbers.



IMPORTANT: pins 9/10/11 = Spark B / Spark C / Spark A.
Pins 21/23/25/27 = Launch / Idle / Nitrous1 / VVT.
Use physical pin numbers, not the order in a copied text column.

05 | Factory Ford connector – cavities 1 to 52

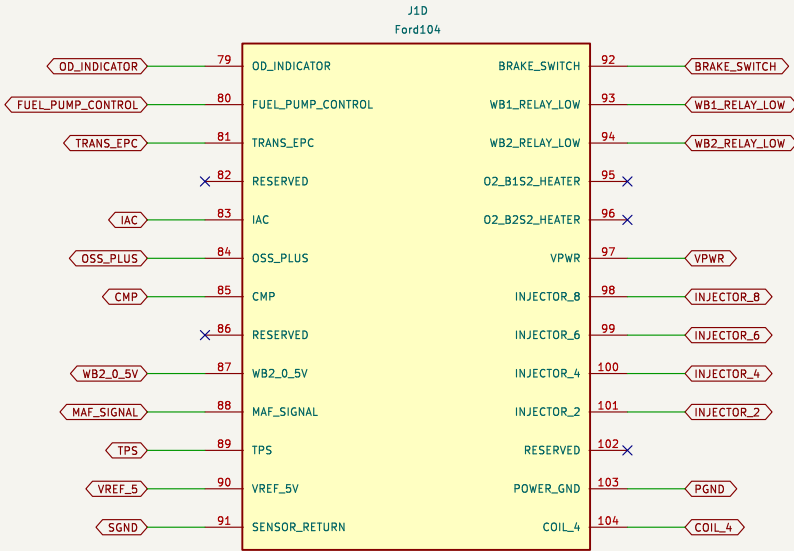
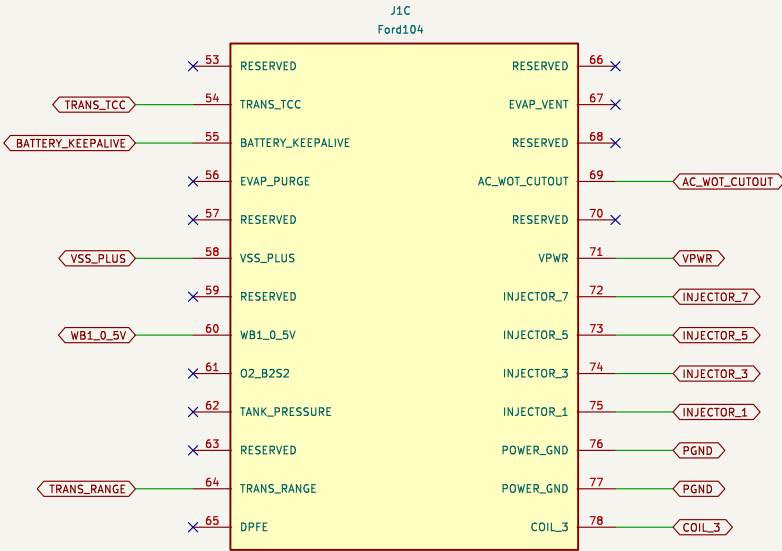
J1 units A–D represent ONE physical salvaged 104–pin connector. Unassigned cavities are intentionally isolated in this draft.



Pin identities: user–accepted Grand Marquis chart, not the earlier Mustang/COP diagrams.
User has confirmed pad numbering agrees with supplied TE reference. Footprint geometry is not assigned here.
Power/ground aliases are deliberate. Coil_1..4 mean factory driver channels, not cylinder numbers.
Service–only/emissions/diagnostic nets continue to the retention sheet; they are not silently declared supported.

06 | Factory Ford connector – cavities 53 to 104

J1 units A–D represent ONE physical salvaged 104–pin connector. Unassigned cavities are intentionally isolated in this draft.

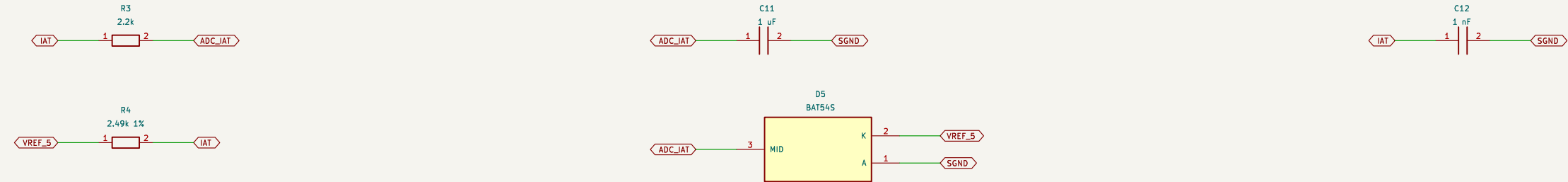


Pin identities: user–accepted Grand Marquis chart, not the earlier Mustang/COP diagrams.
User has confirmed pad numbering agrees with supplied TE reference. Footprint geometry is not assigned here.
Power/ground aliases are deliberate. Coil_1..4 mean factory driver channels, not cylinder numbers.
Service–only/emissions/diagnostic nets continue to the retention sheet; they are not silently declared supported.

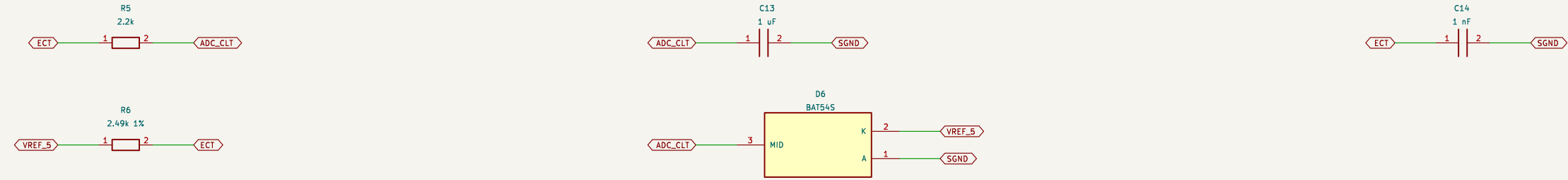
07 | Engine temperature, throttle and oxygen inputs

Each row is one complete input filter. ADC clamps share the sensor–ground and reference domains; thermistor calibration remains required.

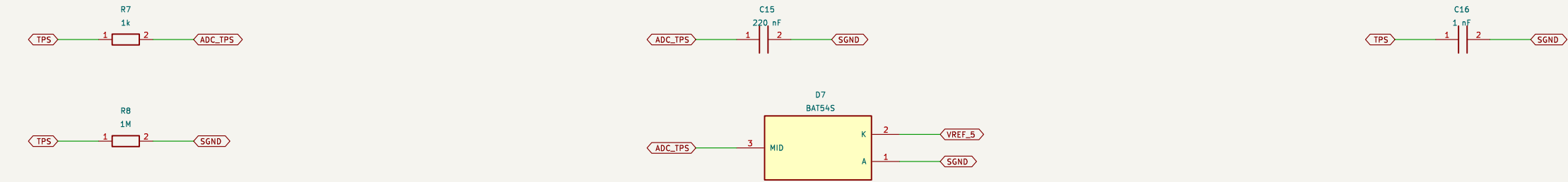
IAT -> ADC_IAT



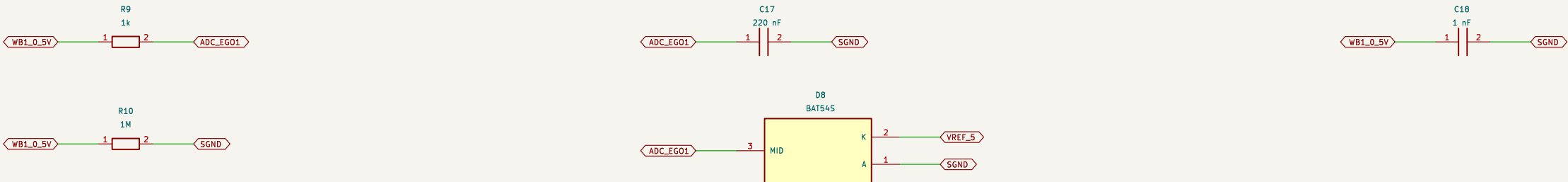
ECT -> ADC_CLT



TPS -> ADC_TPS



O2_B1S1 -> ADC_EGO1

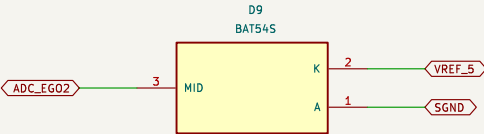
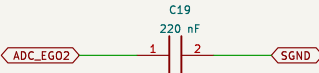
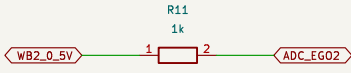


EGO1 is a factory narrowband input. A wideband requires its own controller and correct 0–5 V calibration. Factory TPS remains supplied only by this ECU VREF_5. MicroSquirt obtains TPS over CAN.

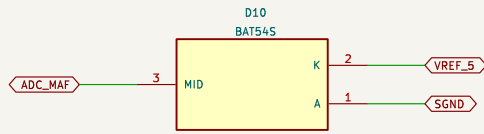
08 | MS3X analogue inputs

Each row is one complete input filter. ADC clamps share the sensor–ground and reference domains; thermistor calibration remains required.

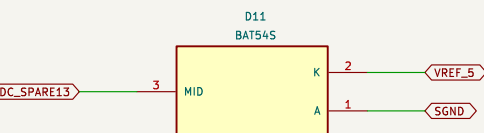
O2_B2S1 -> ADC_EG02



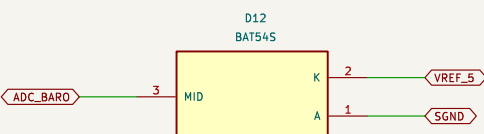
MAF_SIGNAL -> ADC_MAF



AUX_AN13 -> ADC_SPARE13



AUX_AN6 -> ADC_SPARE6



MS3X AN11 is assigned to MAF; AN12 to the second front oxygen sensor.
AN13 and AD6 are available at the expansion connector. Firmware MAF pin selection must match AN11.
Factory rear O2 sensors are parked on the retention header, with no OBD catalyst–monitor implementation.

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1997 Grand Marquis / genuine MS3

Sheet: /08 | MS3X analogue inputs/

File: 08_expansion_analog.kicad_sch

Title: 08 | MS3X analogue inputs

Size: A3 Date: 2026-09-12

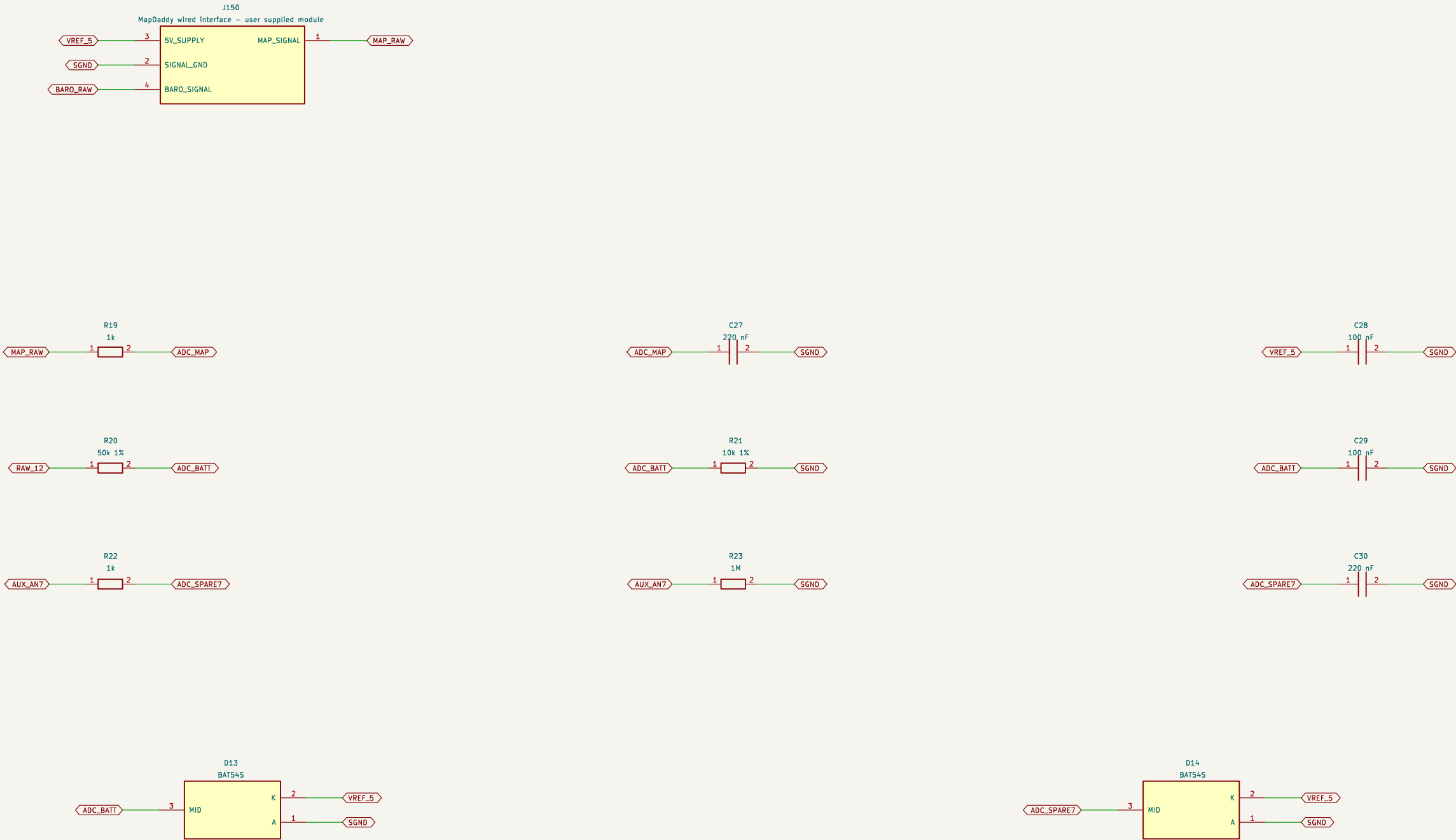
KiCad E.D.A. kicad-cli 7.0.11+dfsg-1build4

Rev: R4 CURRENT-TARGET RE

Id: 9/27

09 | MAP sensor, battery measurement and spare ADC

Internal MAP sensor permits speed-density operation; vacuum hose and package height require mechanical planning.



ADC_BATT measures protected power after series diode; calibrate voltage against a meter.
The analog rail clamps do not constitute a fully validated automotive input protection design.

NOT FOR FABRICATION OR VEHICLE CONNECTION

1997 Grand Marquis / genuine MS3

Sheet: /09 | MAP sensor, battery measurement and spare ADC/
File: 09_map_battery.kicad_sch

Title: 09 | MAP sensor, battery measurement and spare ADC

Size: A3 Date: 2026-09-12

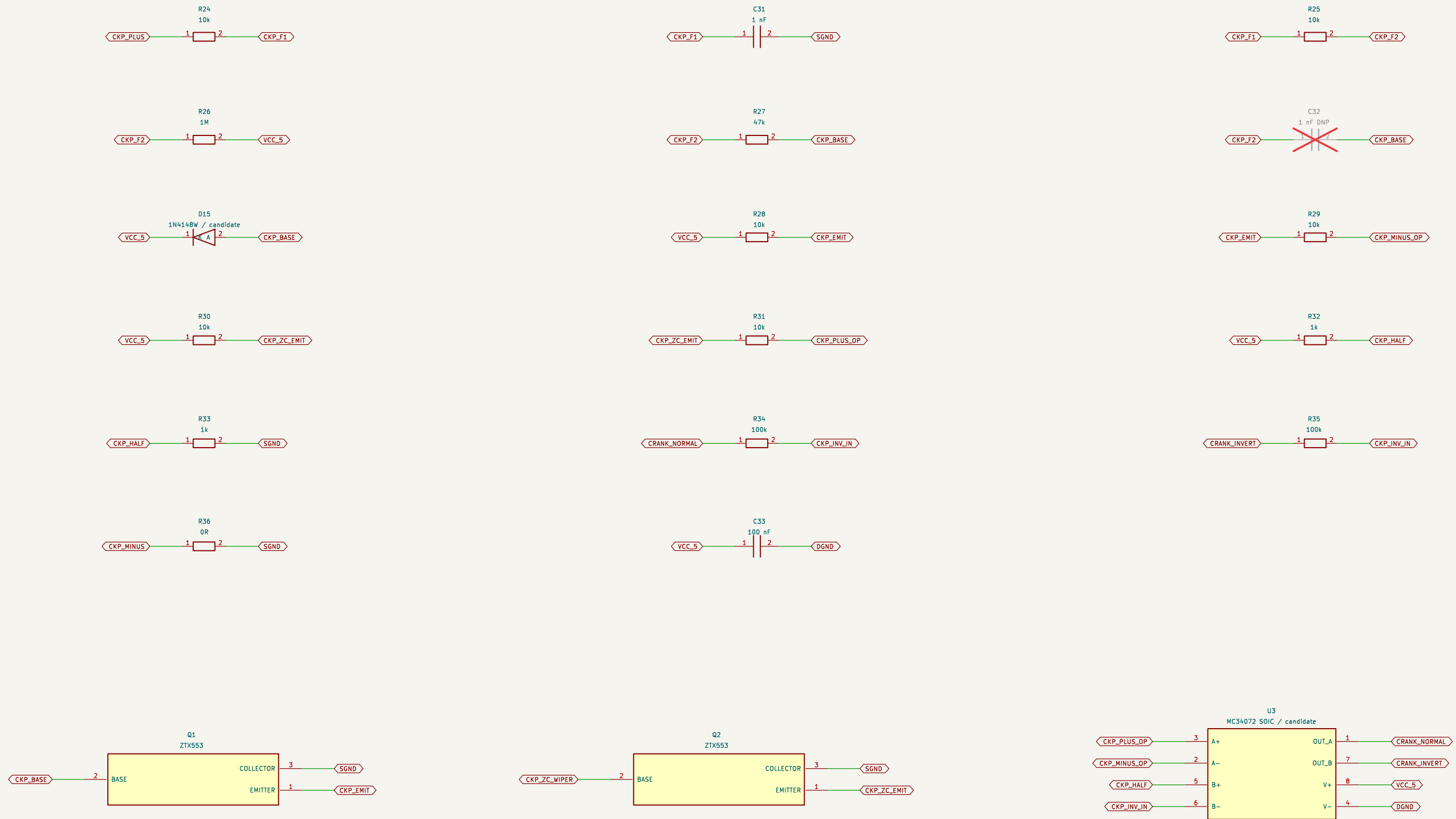
KiCad E.D.A. kicad-cli 7.0.11+dfsg-1build4

Rev: R4 CURRENT-TARGET RE

Id: 10/27

10 | Factory crank VR conditioner

V3.0 zero-crossing circuit. CKP_MINUS returns at sensor ground; phase and hysteresis must be verified with the tooth logger.



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1997 Grand Marquis / genuine MS3

Sheet: /10 | Factory crank VR conditioner/
File: 10_crank_vr.kicad_sch

Title: 10 | Factory crank VR conditioner

Size: A3	Date: 2026-09-12
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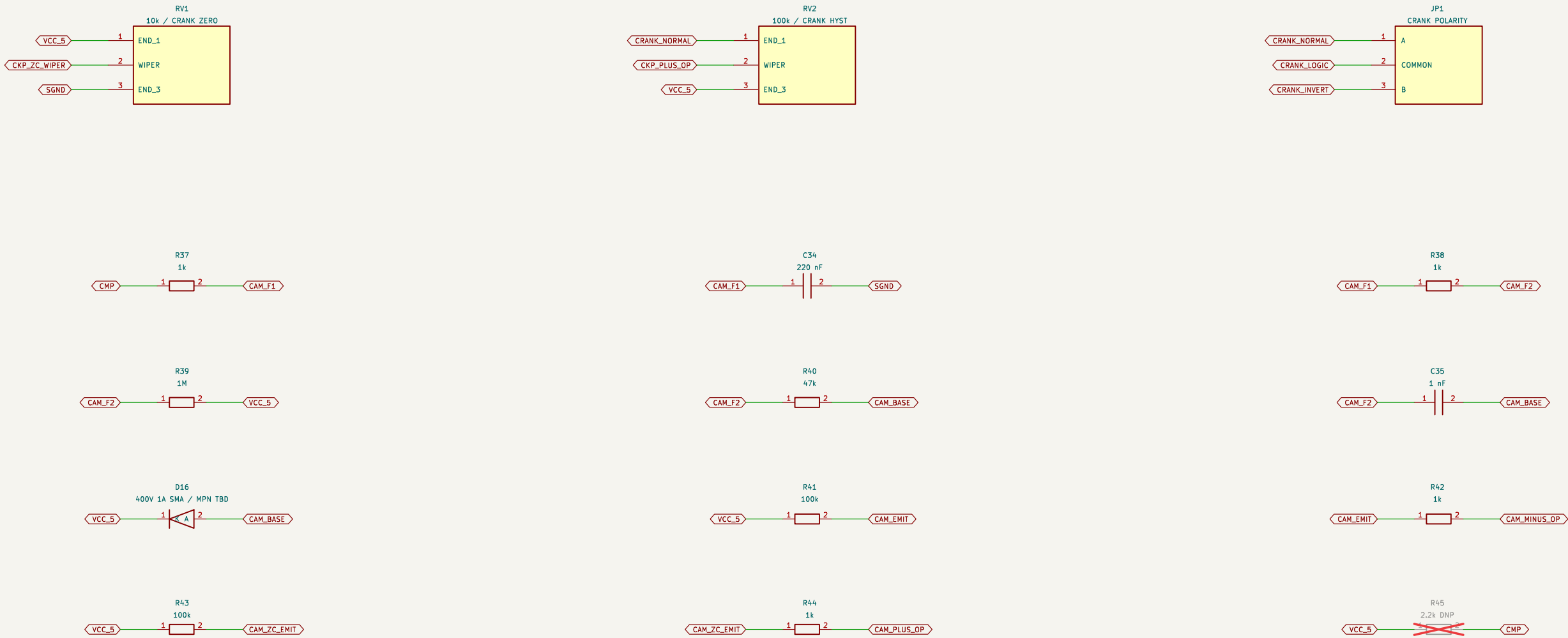
Size: A3	Date: 2020-09-12
KiCad E.D.A. kicad-cli 7.0.11+dfsg-1build4	

Rev: R4 CURRENT-TARGET RE

Id: 11/27

11 | Crank adjustment, phase selection and cam front end

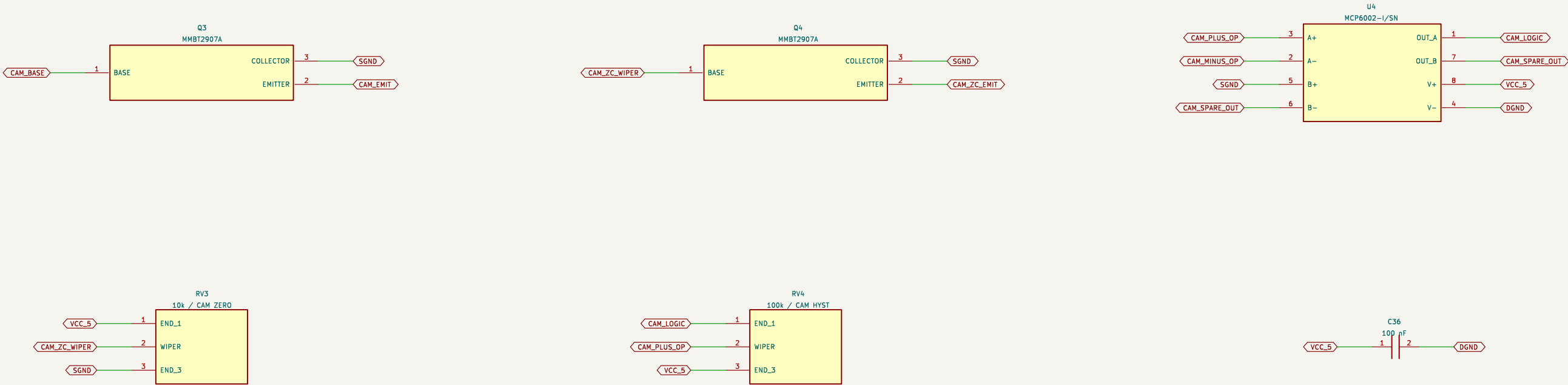
Factory sensor type is retained. Fit only the pull-up appropriate to the actual cam sensor; the factory VR option has no pull-up.



Crank normal/inverted selection is a physical shunt, not two outputs shorted together.
Adjust zero and hysteresis from minimum according to the MS3 setup procedure; settings are not a finished tune.
The V3.0 input assumes a ground-referenced VR return. Confirm actual harness shielding and sensor return.

12 | Cam zero-crossing comparator

MS3X cam-input topology, with the spare op-amp wired as a stable ground follower.



Cam input is module JP2 pin 16 (PT2), not the optional base-socket JS10 input.
Both crank and cam must show clean synchronization from cranking through maximum RPM before enabling loads.
Use a current-limited bench supply and the available JimStim/JimStimX for staged signal checks.

NOT FOR FABRICATION OR VEHICLE CONNECTION

1997 Grand Marquis / genuine MS3

Sheet: /12 | Cam zero-crossing comparator/
File: 12_cam_comparator.kicad_sch

Title: 12 | Cam zero-crossing comparator

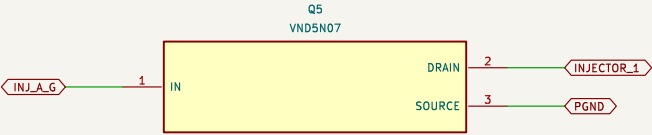
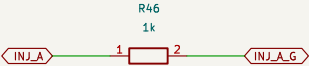
Size: A3 | Date: 2026-09-12
KiCad E.D.A. kicad-cli 7.0.11+dfsg-1build4

Rev: R4 CURRENT-TARGET RE
Id: 13/27

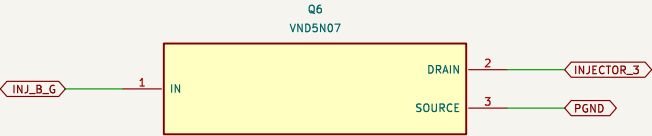
13 | Sequential injectors A–D

VND5N07 protected low-side channels, MS3X functional topology. Stock high-impedance injectors only; no peak-and-hold control.

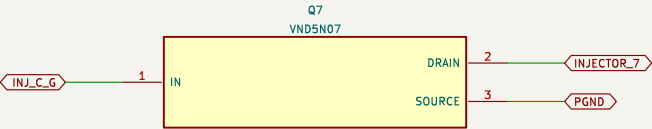
INJ A -> cylinder 1 -> Ford cavity 75



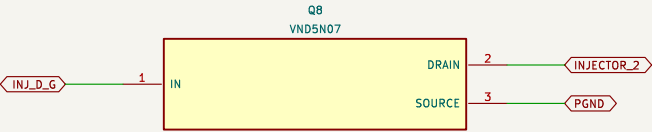
INJ B -> cylinder 3 -> Ford cavity 74



INJ C -> cylinder 7 -> Ford cavity 72



INJ D -> cylinder 2 -> Ford cavity 101



Sequential channel order is planned as 1–3–7–2–6–5–4–8; confirm cylinder identification during output tests.
Drain tab is electrically live. Do not add a diode to +12 across an injector without reviewing closing time.
ST lists this original reference device as obsolete; sourcing or a separately reviewed redesign is required.

NOT FOR FABRICATION OR VEHICLE CONNECTION

1997 Grand Marquis / genuine MS3

Sheet: /13 | Sequential injectors A–D/
File: 13_injectors_1.kicad_sch

Title: 13 | Sequential injectors A–D

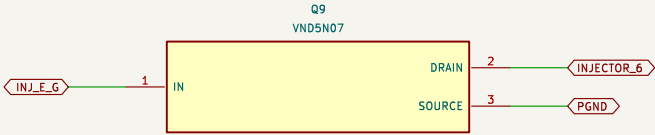
Size: A3 | Date: 2026–09–12
KiCad E.D.A. kicad-cli 7.0.11+dfsg-1build4

Rev: R4 CURRENT-TARGET RE
Id: 14/27

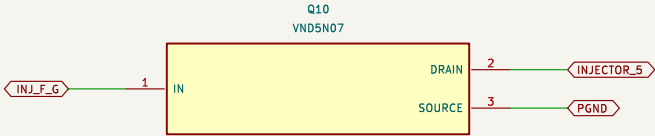
14 | Sequential injectors E–H

VND5N07 protected low-side channels, MS3X functional topology. Stock high-impedance injectors only; no peak-and-hold control.

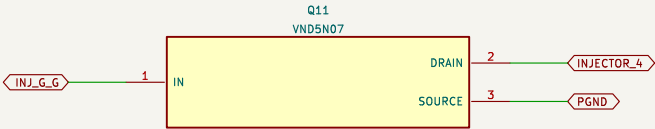
INJ E -> cylinder 6 -> Ford cavity 99



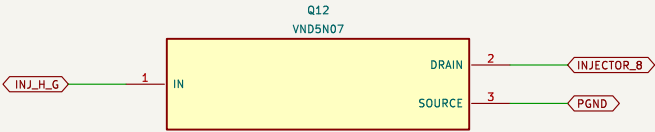
INJ F -> cylinder 5 -> Ford cavity 73



INJ G -> cylinder 4 -> Ford cavity 100



INJ H -> cylinder 8 -> Ford cavity 98



Sequential channel order is planned as 1–3–7–2–6–5–4–8; confirm cylinder identification during output tests.
Drain tab is electrically live. Do not add a diode to +12 across an injector without reviewing closing time.
ST lists this original reference device as obsolete; sourcing or a separately reviewed redesign is required.

NOT FOR FABRICATION OR VEHICLE CONNECTION

1997 Grand Marquis / genuine MS3

Sheet: /14 | Sequential injectors E–H/
File: 14_injectors_2.kicad_sch

Title: 14 | Sequential injectors E–H

Size: A3 | Date: 2026–09–12

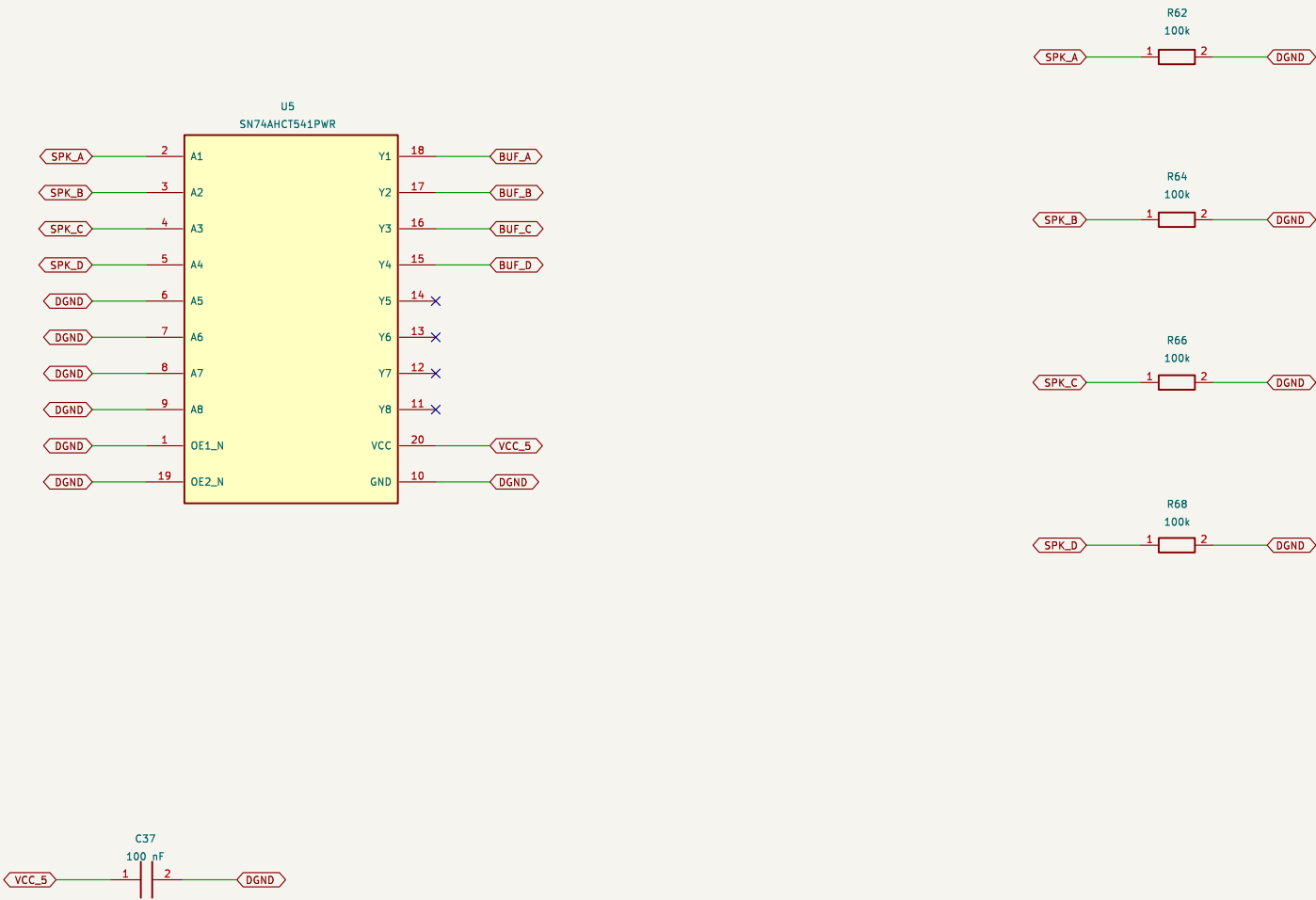
KiCad E.D.A. kicad-cli 7.0.11+dfsg-1build4

Rev: R4 CURRENT-TARGET RE

Id: 15/27

15 | Eight buffered smart-coil signals

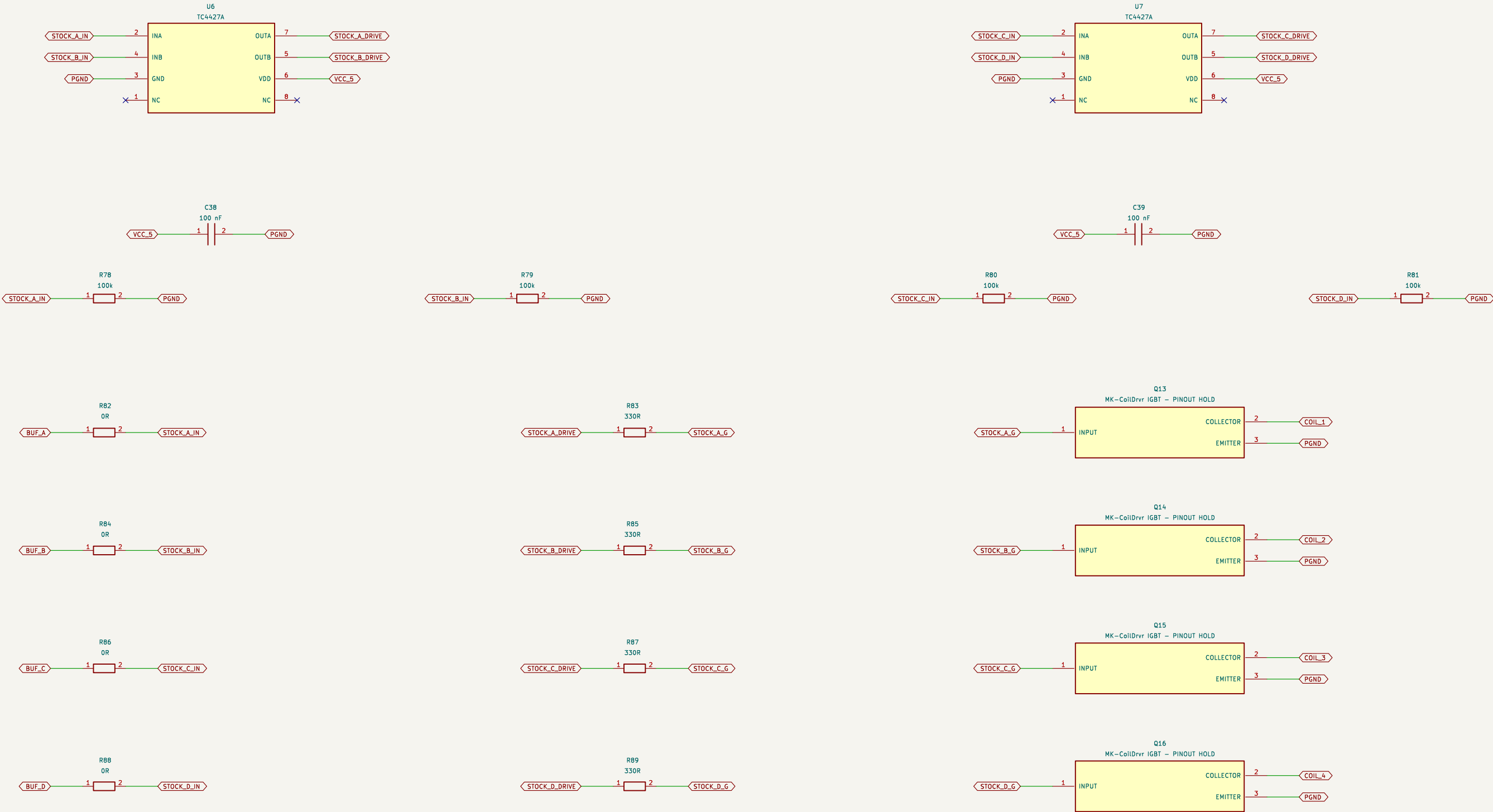
Non-inverting 5 V buffer as on MS3X. D585 coils require a new external coil harness; the stock connector has four ignition channels.



D585 mode: factory power-IGBT inputs disconnected; J110 feeds logic inputs only.
Coil +12 and power grounds are external, fused and returned to the engine.
Select documented spark polarity and measure safe dwell for the exact coils.
Over-dwell self-firing behavior must be checked; no tune is supplied.

16 | Optional factory wasted–spark power stages

Four BIP373 channels. Default population is DNP until current, dwell, driver sourcing and enclosure heat sinking are resolved.

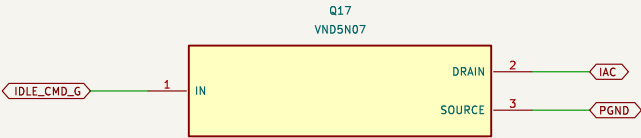


Planned pairs: A=1/6, B=3/5, C=7/4, D=2/8. Verify factory coil–channel wiring before fitting shunts.
Do not connect D585 logic inputs to COIL_1..4: these are high–voltage primary–switch nodes.
This sheet is an alternative population; two ignition modes are not operated together.

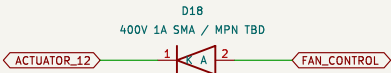
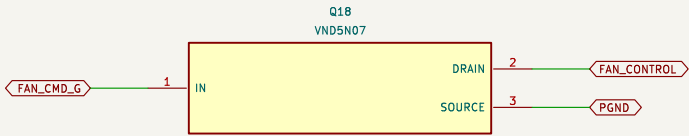
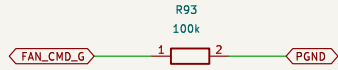
17 | Low-side auxiliary drivers – group 1

Custom protected low-side stages replace the MS3X shared clamp arrangement. Verify each actual load, flyback and PWM requirement.

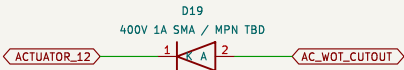
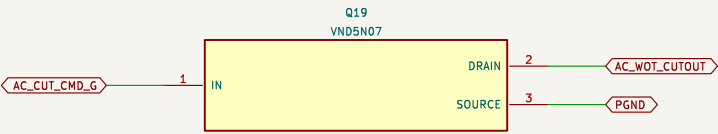
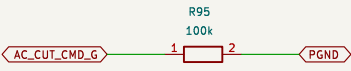
IAC | PWM idle valve



FAN_CONTROL | Fan RELAY only



AC_WOT_CUTOUT | A/C relay logic polarity TBD



ACTUATOR_12 is a separate flyback–return connection to the actual load supply, not regulator RAW_12.
Do not power fan motors, fuel pumps or clutch coils through relay–rated outputs without a load review.
MIL is programmable indication; it does not implement Ford OBD diagnostics.

NOT FOR FABRICATION OR VEHICLE CONNECTION

1997 Grand Marquis / genuine MS3

Sheet: /17 | Low-side auxiliary drivers – group 1/
File: 17_aux_1.kicad_sch

Title: 17 | Low-side auxiliary drivers – group 1

Size: A3 | Date: 2026–09–12

KiCad E.D.A. kicad-cli 7.0.11+dfsg–1build4

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Id: 18/27

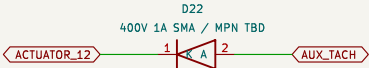
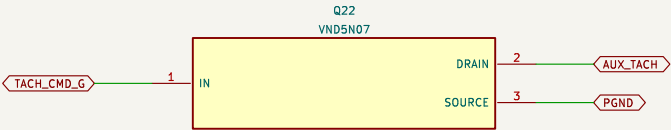
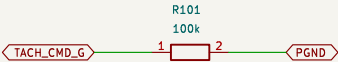
18 | Low-side auxiliary drivers – group 2

Custom protected low-side stages replace the MS3X shared clamp arrangement. Verify each actual load, flyback and PWM requirement.

EVAP_PURGE | Purge valve PWM, firmware assignment TBD

EVAP_VENT | Vent solenoid, firmware assignment TBD

AUX_TACH | Tachometer output



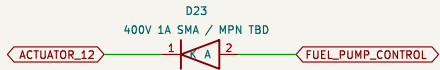
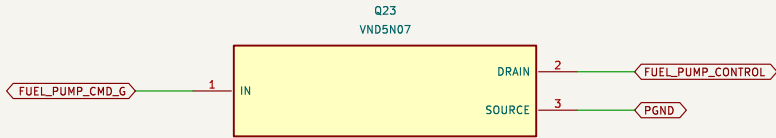
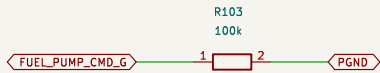
ACTUATOR_12 is a separate flyback–return connection to the actual load supply, not regulator RAW_12. Do not power fan motors, fuel pumps or clutch coils through relay–rated outputs without a load review. MIL is programmable indication; it does not implement Ford OBD diagnostics.

NOT FOR FABRICATION OR VEHICLE CONNECTION		
1997 Grand Marquis / genuine MS3		
Sheet: /18 Low-side auxiliary drivers – group 2/ File: 18_aux_2.kicad_sch		
Title: 18 Low-side auxiliary drivers – group 2		
Size: A3	Date: 2026-09-12	Rev: R4 CURRENT-TARGET RE
KiCad E.D.A. kicad-cli 7.0.11+dfsg-1build4		Id: 19/27

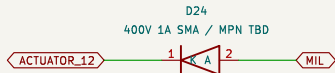
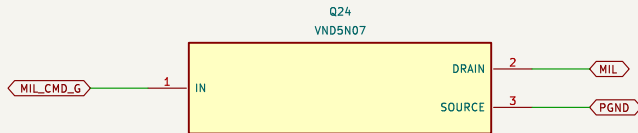
19 | Low-side auxiliary drivers – group 3

Custom protected low-side stages replace the MS3X shared clamp arrangement. Verify each actual load, flyback and PWM requirement.

FUEL_PUMP_CONTROL | Fuel pump RELAY only



MIL | Check-engine lamp, programmable status only



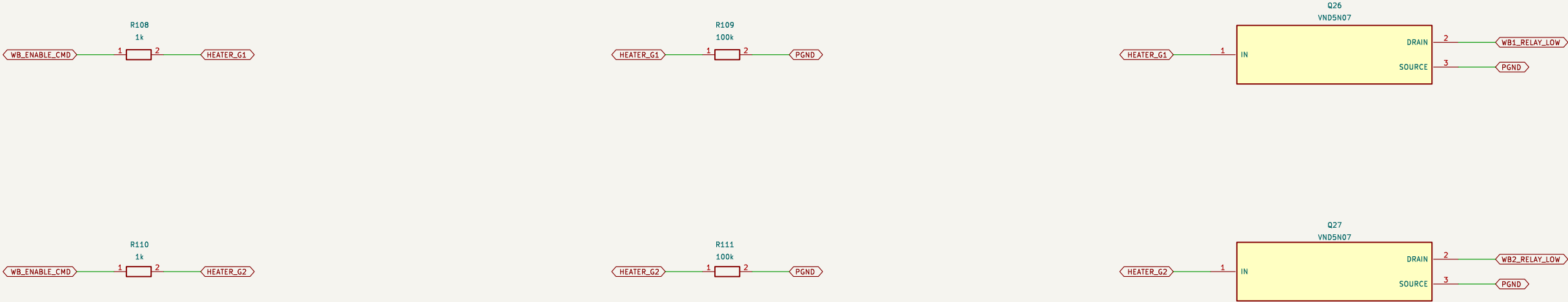
EGR_REGULATOR | Optional EGR valve output, DNP

ACTUATOR_12 is a separate flyback–return connection to the actual load supply, not regulator RAW_12. Do not power fan motors, fuel pumps or clutch coils through relay–rated outputs without a load review. MIL is programmable indication; it does not implement Ford OBD diagnostics.

NOT FOR FABRICATION OR VEHICLE CONNECTION		
1997 Grand Marquis / genuine MS3		
Sheet: /19 Low-side auxiliary drivers – group 3/ File: 19_aux_3.kicad_sch		
Title: 19 Low-side auxiliary drivers – group 3		
Size: A3	Date: 2026-09-12	Rev: R4 CURRENT-TARGET RE
KiCad E.D.A. kicad-cli 7.0.11+dfsg-1build4		Id: 20/27

20 | Wideband-controller relay low-side stages

Two protected low-side devices switch external relay coils; common ON/OFF enable. Do not use heater PWM.

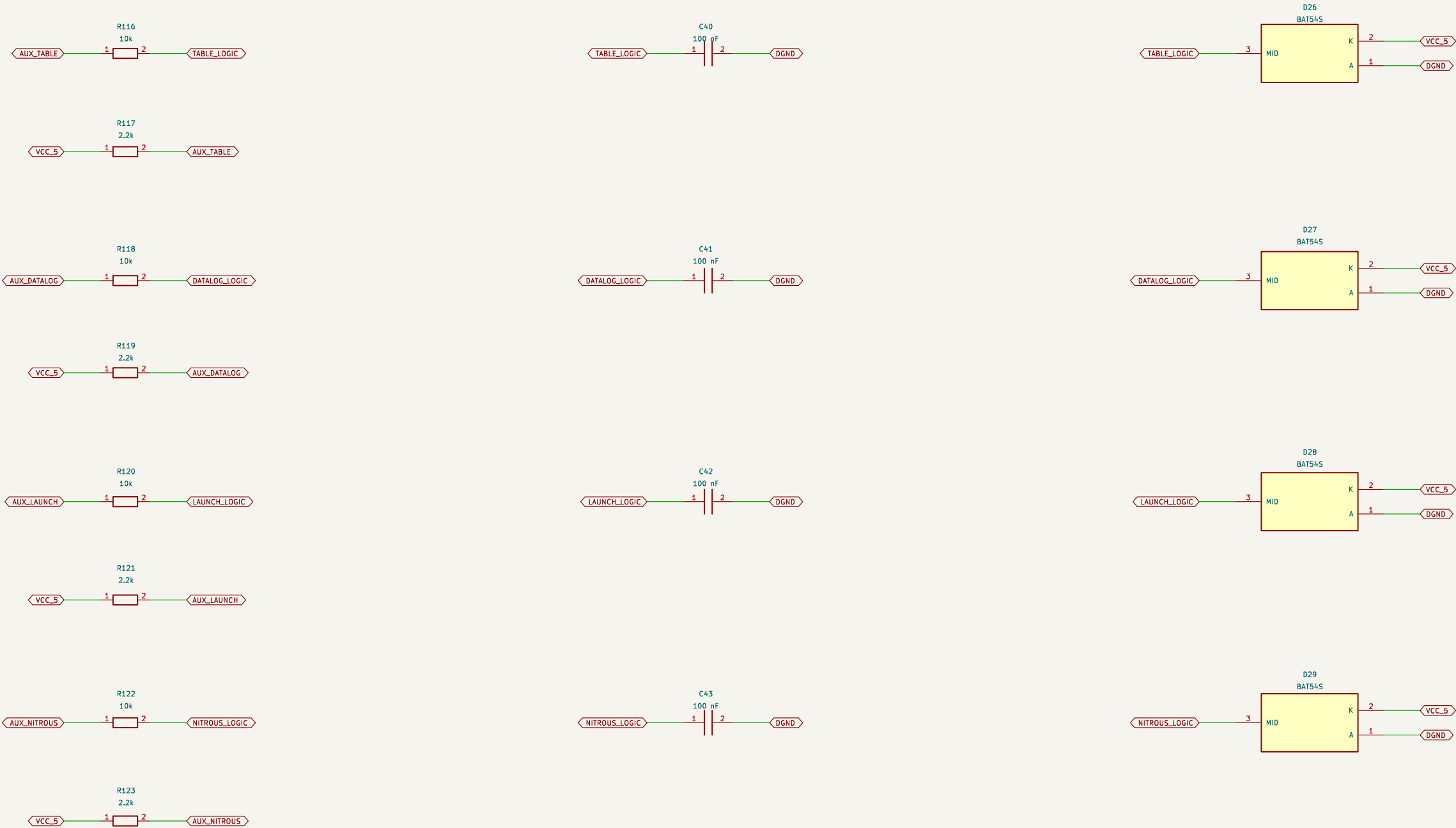


External relay contacts switch supplied fused +12 V to each matched wideband controller.
Controller ground remains connected; 0–5 V outputs feed WB1_0.5V / WB2_0.5V.
Use coil suppression (diode cathode to coil +12 V, anode to the low-side return) and ON/OFF firmware control.
The wideband controller, not this relay stage, regulates the LSU heater.

NOT FOR FABRICATION OR VEHICLE CONNECTION		
1997 Grand Marquis / genuine MS3		
Sheet: /20 Oxygen-sensor heater low-side stages/ File: 20_oxygen_heaters.kicad_sch		
Title: 20 Wideband-controller relay low-side stages		
Size: A3	Date: 2026-09-12	Rev: R4 CURRENT-TARGET RE
KiCad E.D.A. kicad-cli 7.0.11+dfsg-1build4		Id: 21/27

21 | Protected switch inputs

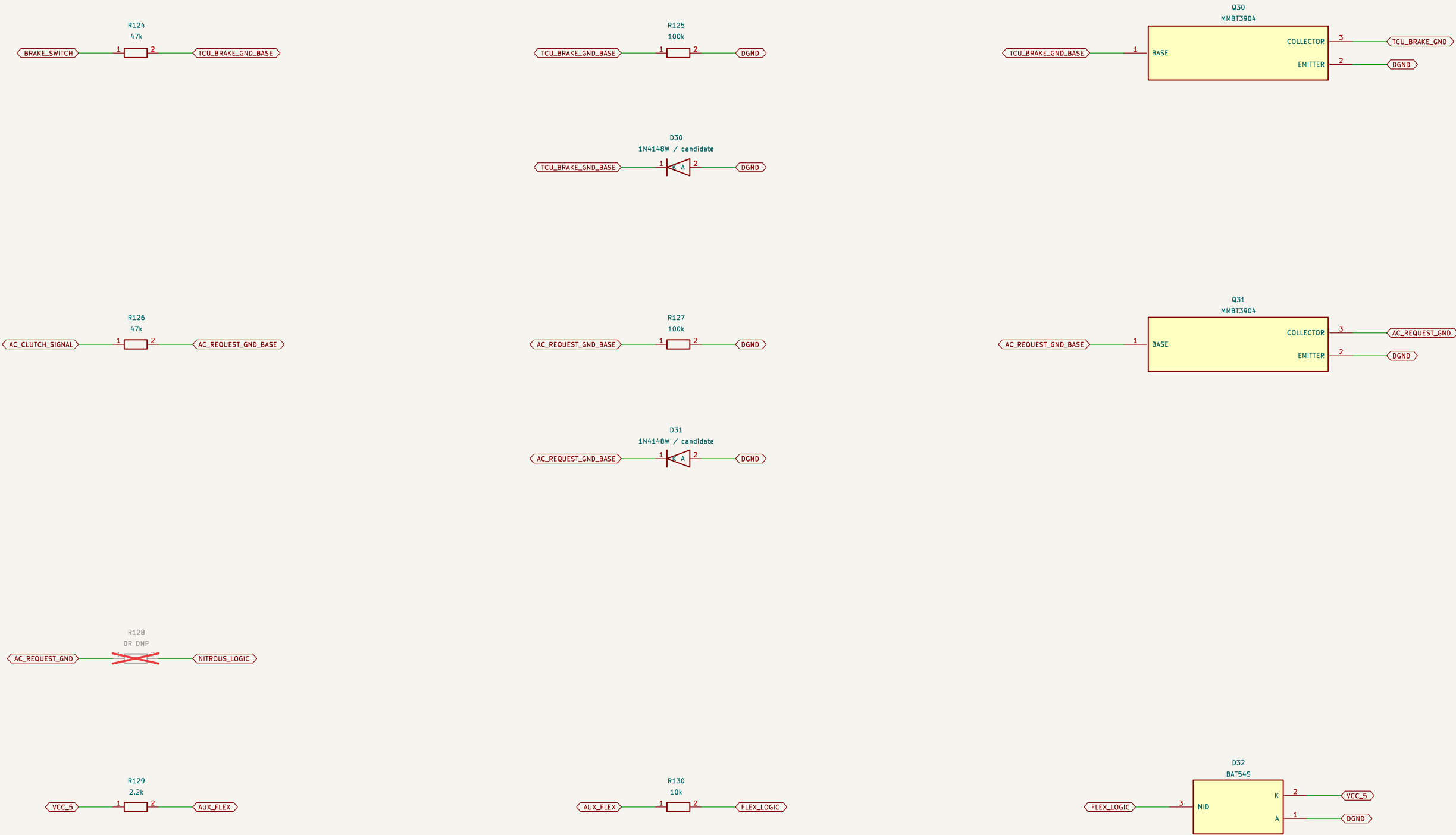
Ground-switching accessory inputs use 5 V pull-ups. High-voltage vehicle inputs use a separate transistor stage.



Input names follow MS3X functions. These connectors do not imply those features are enabled in firmware.
A/C request can use NITROUS_LOGIC by replacing its accessory connection with the conditioned A/C collector.
The same MS3 input cannot be assigned both jobs simultaneously.

22 | Brake, A/C request and flex-fuel input

Brake input is inverted to an open-collector ground signal for MicroSquirt. Positive vehicle inputs do not reach MCU pins directly.

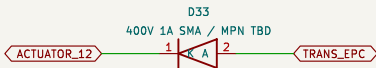
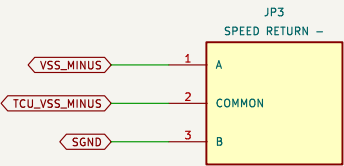
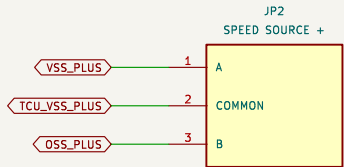
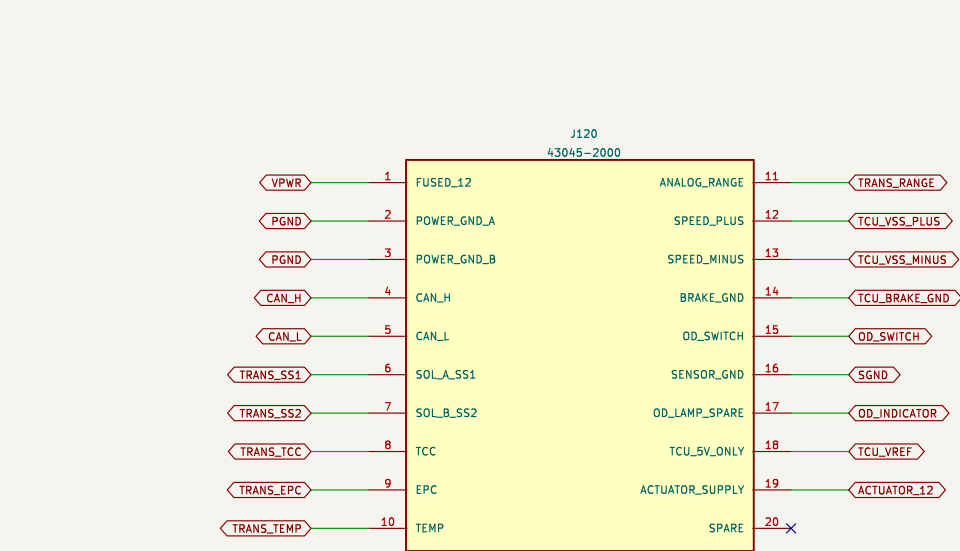


Brake pressed at +12 V -> TCU_BRAKE_GND low. Confirm polarity in transmission firmware.
Flex input has no 100 nF filter to preserve frequency edges; sensor type and waveform need validation.

NOT FOR FABRICATION OR VEHICLE CONNECTION		
1997 Grand Marquis / genuine MS3		
Sheet: /22 Brake, A/C request and flex-fuel input/ File: 22_vehicle_switches.kicad_sch		
Title: 22 Brake, A/C request and flex-fuel input		
Size: A3	Date: 2026-09-12	Rev: R4 CURRENT-TARGET RE
KiCad E.D.A. kicad-cli 7.0.11+dfsg-1build4		Id: 23/27

23 | External MicroSquirt transmission interface

J120 is a NEW custom interconnect, not the MicroSquirt AMPSEAL pin numbering. See the harness table for the two ends.

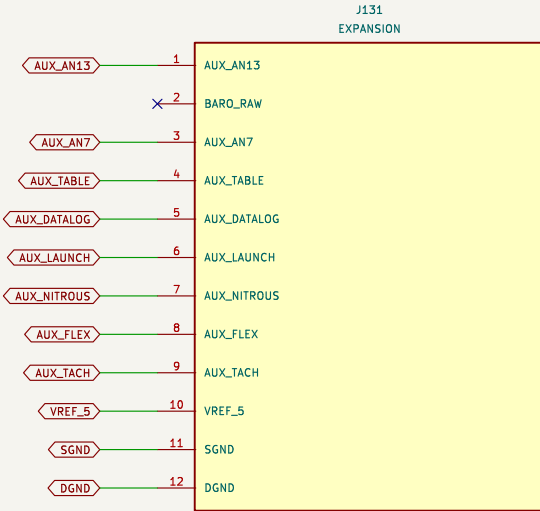
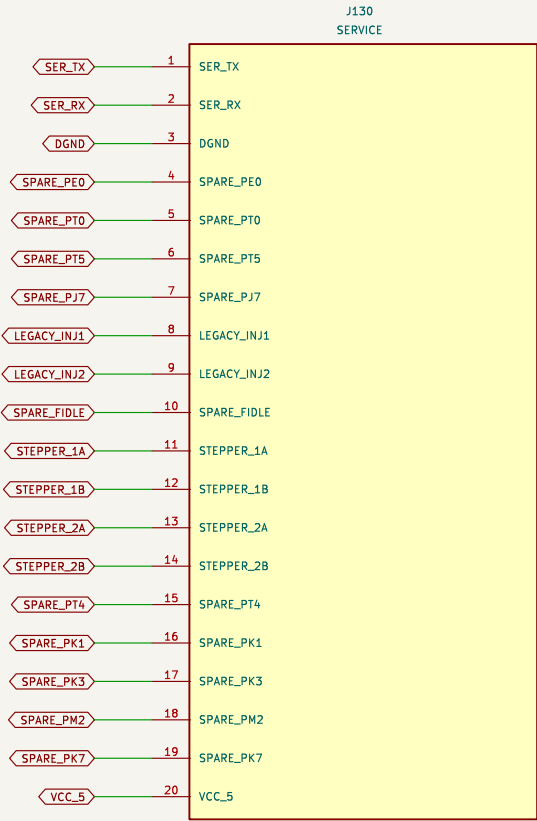


MicroSquirt: RPM / TPS / MAP / CLT over CAN. Do not join its 5 V output to ECU VREF_5.
SPEED jumpers: verify which speed sensor the 1997 transmission provides; select one matched pair only.
J120 pin 17 parks the OD lamp; no verified MicroSquirt output assignment is assumed.
SS1 -> Solenoid A and SS2 -> B are the planned mapping, requiring transmission truth-table confirmation.
Range excitation and calibration cannot be finalized without resistance/voltage measurements.

NOT FOR FABRICATION OR VEHICLE CONNECTION		
1997 Grand Marquis / genuine MS3		
Sheet: /23 External MicroSquirt transmission interface/ File: 23_transmission.kicad_sch		
Title: 23 External MicroSquirt transmission interface		
Size: A3	Date: 2026-09-12	Rev: R4 CURRENT - TARGET RE
KiCad E.D.A. kicad-cli 7.0.11+dfsg-1build4		Id: 24/27

24 | Service and spare feature headers

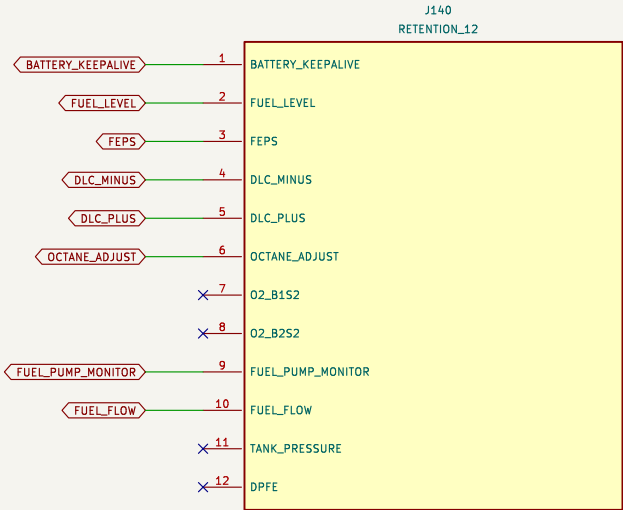
Signal-level expansion only. Legacy injector ports have no driver stages. Genuine MS3 USB and SD are retained on the module.



No extra serial level converter is needed for normal USB tuning on the genuine module.
Do not connect RS232 +/- voltage to SER_RX/SER_TX. These are module-side serial nets.
Stepper outputs may already be power outputs on the genuine card; do not use them as ordinary 5 V GPIO.
This is a replacement carrier with MS3X-equivalent functions, not a pin-for-pin reproduction of every V3.0 option.

25 | Factory functions outside the implemented control scope

These circuits are explicitly parked at an isolated service connector; no hidden connection to a 5 V MCU input is implied.



IMPLEMENTED CIRCUIT SCOPE

- Genuine MS3 40-pin, JP2 and injector interfaces.
- Eight sequential injector channels.
- Eight smart-coil outputs or four optional stock power drivers.
- Crank/cam, analog sensors, switch inputs and auxiliary outputs.
- External MicroSquirt pass-through harness and brake interface.

NOT IMPLEMENTED AS FACTORY EQUIVALENTS

- Ford OBD/J1850 link, FEPS programming and readiness monitors.
- Catalyst, EVAP and EGR diagnostic strategies.
- Factory keep-alive functions and trip/fuel calculations.
- A verified OD lamp command from the transmission controller.

Electrical switches for a valve/heater do not recreate its factory control or diagnostics. No claim of factory functional parity.

UNRESOLVED RELEASE ITEMS

Final connector/IC footprints; full PCB outline and mounting coordinates; current/thermal budget; firmware pin availability and output polarity; transmission range/speed selection; real-load testing. No Gerbers, PCB routing, production BOM or vehicle-ready tune are included in Revision A1.

NOT FOR FABRICATION OR VEHICLE CONNECTION

1997 Grand Marquis / genuine MS3

Sheet: /25 | Factory functions outside the implemented control scope/
File: 25_retention_limits.kicad_sch

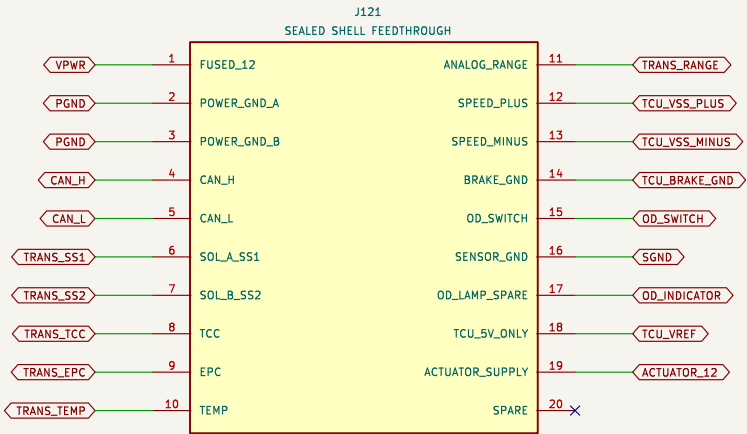
Title: 25 | Factory functions outside the implemented control scope

Size: A3 Date: 2026-09-12
KiCad E.D.A. kicad-cli 7.0.11+dfsg-1build4

Rev: R4 CURRENT-TARGET RE
Id: 26/27

26 | ECM–shell pass–through to cabin MicroSquirt

J120 is on the PCB. J121 is a sealed shell–mounted connector, joined by a short internal pigtail.



PHYSICAL HARNESS PATH
Factory transmission wiring -> Ford J1 -> new PCB J120
-> short internal pigtail -> shell J121 -> cabin cable
-> MicroSquirt V3 35–way AMPSEAL.

J120 cavity n mates to J121 cavity n electrically.
MicroSquirt cavity numbers DIFFER: use the harness table.
J121 is off–board; omit from PCB placement.

The cabin cable carries solenoid CURRENT as well as
sensor signals, power/returns and CAN. It is not a
CAN–only or ribbon–cable connection.

Twist CAN H/L together; route VR speed as a pair.
Keep solenoid PWM currents out of sensor–return wiring.
Provide strain relief and a sealed cable route.
A shell opening is distinct from a firewall opening.

J121 contact rating, conductor sizes, terminal seals, cable length and shell cutout are not yet selected.
Pins 17, 19 and 20 have no assigned MicroSquirt signal endpoint; isolate/seal unused cabin contacts.
Mount the complete MicroSquirt inside the car, with its tuning connector accessible.
This change preserves the planned factory connector routes; it does not resolve the prior range/speed/ground review items.